Hardware Determinism over Statistical Gradients:

A GPU-Accelerated Boolean Attack on the Final U-534
Enigma M4 Ciphertext (P1030680)

Digital Defiance
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Abstract

For decades, message P1030680 has remained the sole unbroken Enigma M4 transmission from the May 1, 1945 intercept of German submarine U-534. Modern attempts to decrypt this 72-letter ciphertext have historically relied on statistical hill-climbing and distributed computing. This report shows that purely statistical attacks fail against short M4 ciphertexts due to plugboard overfitting, documents a mid-message middle-rotor turnover flaw in whole-message trigram scoring, and describes a software Welchman diagonal board on Apple Silicon Metal (40 million machine settings per second) with a mandatory $\leq$ 10-plug completion kill chain. We record graded controls (known-message break), archival eliminations, clean negatives under catalog and operational cribs, a quarantine path for soft near-misses, and a parallel TensorLUT continuous–discrete compiler track. A runtime benchmark selected a four-lane unified Verilog Future Bank. The unified P1030680 Mulein search machine and overall campaign carry the formal name *Nazi Blaster 9000*; its historical 261-entry canonical-menu/Future campaign is *Fahrenheit 261*. That historical name denotes the inventory and campaign, not executed breadth: its bounded launch ran only identity Future 0 at settings $0 \leq s < 256$, leaving 260 Futures unexecuted. A separate 628-Future operational-prior preflight then covered setting 0 across all identity and post-gap entries: 16,328 checked receipts produced 15 host-replayed one-edge repaired physical candidates, all non-exact, with zero exact hits and zero BREAK gates. The same shell and all 628 Futures then completed settings $1 \leq s < 256$: 10,048 synchronized chunks and 4,163,640 checked receipt projections retained 4,184 host-verified physical candidates but fired zero BREAK gates. Forty exact identity hits died at $\leq$ 10-plug completion; repaired identity and all post-gap candidates still require correction or geometry-aware full-message replay, and none supplied a scored plaintext, key, or decrypt. This is bounded evidence, not global coverage or evidence that the ciphertext is garbled. **We do not claim a decrypt of P1030680.** Campaign fitness is cleartext Metal batch, not HELUT's encrypted torus path (documented separately in `paper/helut.tex`).

1 The Failure of Statistical Gradients on Short Ciphertexts

When attempting to break an Enigma ciphertext, modern cryptographic approaches default to statistical optimization, such as Index of Coincidence (IC), n-gram fitness, and genetic algorithms. However, empirical testing against P1030680 proves that statistics cannot *search* a 72-letter Enigma space; they can only *rank* a small list of survivors.

The Enigma plugboard (the *Steckerbrett*) offers roughly 47 bits of freedom. On a short message, this immense freedom causes statistical models to overfit. Algorithms will output high-scoring strings of letters that closely resemble German syntax, but these strings are mathematical hallucinations. Control tests on a known, solved message (P1030684) confirmed that purely statistical engines favor these false positives over the true key. Consequently, any successful attack on short M4 intercepts must abandon statistical guessing in favor of absolute Boolean logic.

2 Emulating the Welchman Diagonal Board in GPU Shaders

To execute a deterministic attack, the hardware architecture of the 1945 Enigma must be perfectly mapped to modern hardware. Using Homomorphic Edge Look-Up Tensors (HELUT), we engineered a digital Bombe inside Apple Silicon Metal shaders.

This software-defined FPGA evaluates Gordon Welchman's diagonal board graph reductions at a rate of 40 million machine settings per second. The engine utilizes Turing-shaped reductions based on the physical reality of the short-message M4:

- Because the far-left Enigma rotor (the Greek wheel) never steps during a short message, its ring can be digitally pinned.
- Because the left rotor's notch drives nothing, its ring can also be pinned.
- Together, this physical constraint collapses the total search space by a massive factor of 676.

The mathematical soundness of this reduction was verified in a blind control run. Given a 27-letter crib from a known message (P1030684) and no other parameters, the GPU pipeline evaluated 16 billion settings and extracted the true key, exact shell, and all ten historical plugboard cables in just 361 seconds (claim C2). The archived log (`logs/control-p1030684-rings.log`) reads 427 s: 361 s was the first run, and the log was overwritten by a re-run after head scoring was added. Both figures are recorded in `BREAK_P1030680.md`; the re-run is the reproducible one.

3 Physical Constraints vs. Mathematical Ghosts

Translating mathematical graph theory into physical 1945 hardware exposed a significant vulnerability in pure Boolean reductions. Certain crib wedges produce valid Welchman states that are mathematically flawless but physically impossible.

We classify these false positives as "ghosts". In a trial run, a crib of `UUUVIRSIBENNULEINS` at offset 0 produced 12 mathematical survivors out of the Welchman board. However, when mapping the remaining letters, none of these 12 settings could be completed within the physical limit of exactly 10 plugboard cables. Forensics revealed these were "split menus," where connected components within the graph admitted zero valid starting seeds.

To prevent these ghosts from overwhelming the CPU, we built a SAT-completion kill chain inline as the GPU drains. This ensures that only mathematically valid states that *also* fit physical hardware constraints survive the hardware pipeline (claim C3).

4 The Mid-Message Turnover Scoring Flaw

Our research uncovered a systemic flaw in how distributed computing projects historically score Enigma decrypts, likely explaining the decades-long failure to crack P1030680.

Standard algorithmic sweeps utilize a linguistic scanner to score the full 72-letter plaintext. However, if the true key clicks the middle rotor over *after* the crib phrase but *before* the end of the message, the plaintext will be perfectly decrypted across the crib span, but turn to garbage for the remainder of the message.

Because traditional trigram discriminators score the *entire* message, the garbled tail drags the overall score down to the midpoint between German and random noise. This causes the engine to discard the true key, actively rejecting the correct answer.

To rectify this, we designed a **windowed discriminator score**. This metric evaluates the readable head of a decrypt (requiring at least 16 non-crib letters) independently from the whole message, passing the key if either the full message or the head meets the -3.600 threshold. Applying this fix revealed 2,219 physically valid ten-plug boards that had been previously rejected by the flawed whole-message scoring (claim C12).

5 The M-Thetis / Potsdam Register Mismatch

Cryptanalysis is limited by the quality of the plaintext hypotheses (cribs) used as wedges. Archival auditing of the May 1, 1945 U-534 corpus highlighted a severe historical blindspot regarding the target message.

The scraped corpus contains 48 broken M4 messages. Crucially, 31 originate from the Potsdam net, 16 from a second net, and 1 from a third. P1030680 is the only message in the scrape with the working assignment M-Thetis. Kenngruppen *table selection* for 1 May 1945 is now documentary: Tauschtafelplan Bruno, Prüfnr. 1772a, Kennwort Quelle, column six annotated Mai 45, day 1 printed A (Hörenberg scan). That is which Tauschtafel letter to use. Quelle Tafel A pair contents are now photographed in Crypto Museum DoppelTafeln_2499.pdf pp. 3–4 (Prüfnr. 2499, Phase 61.3); the 2499 cover assigns that Ausgabe's Tauschtafelplan the same Prüfnr., so 1772a vs 2499 edition compatibility is not demonstrated, and this is still not a dated Zuteilungsliste. The assignment to M-Thetis remains supported by `ACH` under the later allocation list used by Hörenberg. Phase 75 does not change that layout. Seven of the 48 row-1 groups double at positions 3–4 (chance about 0.002 for that pair). The published rule puts the null first: Hörenberg's Kenngruppen page cites Kenngruppenbuch page IX for skipping that letter, and Erskine's Kenngruppenbuch note gives the same rule. Drop-last would read `EAC` and is not adopted. The public photograph of Zuteilungsliste Teil B prints the filler as 1. Stelle and 4. Stelle. That sheet is the undated 621–653 crop, not a daily key. The Crypto Museum scan of the 1941 Kenngruppenbuch (M.Dv.Nr. 98, Prüf-Nr. 992) is a different allocation sheet, Kennwort Hering, and is not a 1 May key. Column 645, row 14 of that scan reads `ACH`, the same cell the detector already uses. The raw OCR of columns 641–650 is in `Fixtures/kbuch_1941_992_page33_ocr.json`; the other tokens are not adopted. That does not date the net. The other two detector entries match the same book: column 12 row 9 is `DUZ`, and column 211 row 16 is `ZPT` (`Fixtures/kbuch_1941_992_kbook_cells.json`). That is the whole detector table. It does not open a menu. Gallica, the Deutsche Nationalbibliothek, LIBRIS, and the Norwegian Nasjonalbibliotek return no record of that book or of ZTPG 369139. Discovery does not name a piece for ZTPG 369139; the last indexed ZTPG 36 range is HW 18/21, 11–19 April 1945, serials 361317–363999 (DEFE 3/558–560). A reported reading of Tafeln B–H for this indicator is not loaded, and 3 May stays unencoded. P1030710 is U3030 on the Potsdam key, and P1030680 is already excluded from that key at every message key, so traffic inside that cluster would have been put on a different key the same day. The catalogue chronology is not fully settled: Selm Merel Wenselaers reports an earlier list, *Forelle*, which maps column 645 to Aegir while the later list maps it to Thetis. Changing Zuteilungslisten can reconcile the apparent contradiction without changing the K-Buch column; the exact date, codename, and provenance of the later list remain pending. Likewise, preservation among U-534's radio papers establishes recovery provenance, not intended recipient. An intercepted training-network signal is an open hypothesis, not a finding.

Because Bletchley Park never bothered to work the M-Thetis training net, and no other Thetis messages survive in the corpus, all 100 historical cribs used to attack P1030680 were register language borrowed from other operational networks. If Thetis operators opened their 72-letter messages with callsigns or key groups before the body—unlike Potsdam operators—the exact crib placements used in modern attacks are fundamentally the wrong shape.

That reading is correct about *provenance* and has been applied too broadly, which is worth stating because it changes where compute should go. Classifying the 868 tested placements of length $\geq$16 by content rather than by source, **736** contain only universal Kriegsmarine procedure—spelled numerals, `VVV`, `UUU`, `FLOTTX`, `VONVON`—and just **132** carry net-specific addressing such as `KOMXADM`, `CHEF`, or place names. Spelled numerals and procedure markers are service-wide conventions, not Potsdam idiolect, and the most connected of them concern *U-boat flotilla movements in the Baltic*, which is precisely what a Baltic training net existed to coordinate. The

register objection therefore bites hard on openings and addressing and much more weakly on numerals and procedure. **288** of those universal-only placements are single-component with $\geq$8 loops and have never received middle-ring coverage, so a plausible-crib hypothesis remains live in *untested key space* rather than eliminated. The resilience of P1030680 is plausibly a mixture of mismatched operational doctrine and incomplete ring coverage, rather than the cryptographic strength of the cipher itself.

The archival side of that question is now active team work. Selm Merel Wenselaers, a historian and curator in Amsterdam and Antwerp, leads the campaign's historical/archival arm alongside HELUT's mechanical cryptanalysis. Her named DEFE and U-3521 KTB findings report a concrete process in which scuttling parties remained aboard while crews transferred to Neustadt; `NEUSTADT` and `VERSENKUNG` are therefore linked stages rather than competing scenarios. She also reports scheduled `FT-Programmzeit` reception during AGRU-Front training and has enquiries open with archives and collections for radio papers, key material, and a second Thetis message. These reports move priors but remain source-reported until scans or exact transcriptions enter the campaign receipts. Her U-3521 source dossier, U-534 last-days reconstruction, NID 1/PW/REP/17 interrogation copy, and the Quelle Prüfnr. 2499 booklet are now attached on the public campaign journal with her credit. Direct quotations from the dossier's own refusals are used there in preference to paraphrase; email rankings that are not on file are labelled as paraphrase. None of those files mint a crib, identify Thetis, or imply a decrypt.

One associated radio finding is independently reproducible in the repository: P1030668 contains `FFFTTTBLEIBTBESETZT` exactly, while P1030707 carries the corrupted `FFFTTTBLEIBTBESEOZTX`. The clean phrase therefore does not depend on reconstructing the damaged sister copy. Its 19-letter core is a weak menu, but the exact 40-letter P1030668 source context yields four single-component, 18-loop alignment hypotheses. They are a **clean negative** (Phase 60) under the same VIII-fast middle-ring coverage that came back empty on the strongest catalog menus: 4/4 dead at the board, **0 raw stops over** 2.076×10^{11} **settings** at 52.3M/s, 0 quarantine. Scope remains 42 of 336 wheel orders, not wheel-order coverage. The clean U-3024/U-2538 number strings are withdrawn after raw-copy rechecking, and short historical terms are kept independent rather than concatenated into invented sentences.

The findings are joint work: historical research defines the source-grounded questions, and the graded machinery turns them into bounded measurements. A 2026-09-24 archival map (Phase 61) names the documents that would change the search — NARA RG 457 key logs, a reported R.I.P. 401 non-random wheel-order section, and original Kenngruppen from the 3 May Ausbildung/Front split — and has not yet read the NARA boxes. Tauschtafelplan Bruno, Prüfnr. 1772a (Phase 61.1) does date P1030680's Kenngruppen table as Quelle / Tafel A on 1 May; that is table selection. Tafel A pairs are photographed in Prüfnr. 2499 (Phase 61.3); 1772a vs 2499 edition compatibility is open, and this is not a dated Thetis Zuteilungsliste. Bauer's *Ultra versus Enigma* (Phase 61.2) cites Anhang F for Hydra May 1945 Steckerverbindungen, in use through 7 May; the appendix page is not in the chapter extract, is not a Thetis key, and does not by itself answer cross-net Walzenlage assignment. VIII-fast remains an ordering prior. The April 1945 NID interrogation reconstructs Baltic working-up and radio-watch practice but does not identify Thetis, and is not a crib source.

Potsdam and Plaice daily keys for 1 May were eliminated by exhaustion against P1030680 (claim C11): all 26^4 message keys land at the noise floor.

6 Graded Campaign Status (August 2026)

The operational ledger is `BREAK_P1030680.md`; the public claim freeze is `directives/claim-sheet.md`. Table 1 summarizes what is *proven*, what is *eliminated*, and what remains open. **None of these rows is a decrypt of P1030680** (non-claim N5).

Table 1: Campaign grades (cleartext Metal). Fitness is not HELUT encrypted tick rate (N6).

Result	**Status**
Welchman blind control (P1030684)	Break in 361 s first run / 427 s archived re-run; all 10 plugs
Stochastic KPA datapath	Control oracle + near-miss PASS; blind stecker ∼22/72
Training / collapse / weather / RIGA priors	0 survivors under 80% rigor (ceiling ∼60–69% = coincidence)
Exact ≥16 catalog × rings AAAA	Clean negative
Curated exact / fuzzed top-40 × right rings	Clean negative
UEBUNG / Thetis-register arms	AAAA clean negative; register rings **done, 73/73, no break** (Phase 71). Menus 66–73 best IC 0.038, tail −4.629. Menu 62 head misses the whole message. Three `FUNKSPRUCHUEBUNG` quarantine rows are prefix hits; whole-message tails −5.256, −4.534, −4.733 miss the soft band. A naval-register trigram scores those decrypts at 5.4–6.0 bits/letter, and annealing a fresh board on the 55 UEBUNG near-misses and every logged physical rotor setting (235 settings) reaches a best of 4.069, against a held-out naval band of 3.14–3.33 (dense median 4.36). The catalog sweep scores with the dense trigram. Catalog from 418 is **done, no break** under VIII-fast (451/451, best physical IC 0.041, tail −4.488, 42 of 336 orders)
Soft-tail UEBUNG quarantine escalate	55 seeds; Hybrid 0 survivors
Regenbogen / Hannibal legacy Welchman slices	Not BREAK where tested under AAAA/right-ring scope; soft escalate 0 survivors. This does not close correction-aware operational Futures
Own-orders / filler × AAAA	Own: 0 physical; filler below soft — no rings
Catalog exact × right rings	Full orders parked at 417/2513. VIII-fast continuation **done, no break**, 451/451, exited 2026-09-25T23:57:03Z. 642,562,788 raw stops over 9.002×10^{11} settings. Best physical IC 0.041, tail −4.488. Two prefix quarantines on `FFFTTTBLEIBTBESE@17` miss the whole-message soft band. 42 of 336 orders
Crib catalog is recurrence-filtered	**Known but previously unrun surface.** An internal 2026-08-16 relay audit measured **152,938 of 154,469** distinct 16–40 letter corpus windows (**99.0 %**) as one-carrier phrases and expanded them to **2,691,455** legal placements; its 24-placement fixture was never run or printed publicly. Selm Merel Wenselaers (Historian / Curator, Amsterdam / Antwerp) independently flagged the filter and supplied three exact historical leads on 2026-09-22

Selm-led historical/cryptanalytic arm

Joint campaign work. Selm Merel Wenselaers leads the active archival arm; her source research and menu judgement define evidence-backed questions, and HELUT's graded path turns them into controlled measurements. Exact and post-gap tests of her first three leads are both clean 28/28 negatives. A new exact P1030668 `BLEIBTBESETZT` context fixture is a **clean negative** (Phase 60) under VIII-fast × middle ring: 4/4 dead, **0 raw stops over** 2.076×10^{11} **settings**, 0 quarantine. VIII-fast remains an ordering prior pending a cross-net Schlüsseltafel answer (Phase 61: NARA key-log request acknowledged, no box read; Bauer Anhang F is now the linked Hellschreiber scan, Hydra May 1945 stecker, not this board; Tafel D is not applied to the indicators). `FORT` is four letters and is not a crib. ARGUS v3.1 records that its crib export had been only the Hörenberg scrape; the other family in that bundle is a 1943 U-466 KTB transcription and a Nixe sheet, and it is not queued. ARGUS v3.2 still offered a crib command and is superseded by v4, which emits no crib (121 candidates, all ineligible). ARGUS v5 (27 September 2026) adds an open historical branch: U3030, U3032, and U2540 on 1 May, with a reported 18:00 reunion (ZTPG 369139). Phase 76 withdraws the v5 grid ROT07A / AO7819. Her reading of the card is Gedser at 1630 and anchored at 1800 at 7A; the next card distinguishes 7A from ROT 07. TICOM I-38 §31, her reading of the 21 June 1945 interrogation, is the December 1944 decision that only two-turnover wheels go in the right-hand position. The report does not name Thetis and does not select VIII. That is the existing two-notch ordering prior, 126 of 336. The prior does not restrict. Phase 77 records Dean Annison's 1 October 2026 reply: the only searchable Thetis card at Bletchley Park is a Naval Section VI technical-intelligence index card, and he found no `ACH`. Selm reads it as July 1944 with the cross-reference `F.M.T. 2 (q.v.)`. That string matches the published designation of the Thetis IIC radar decoy (FuMT 2), not Schlüsselbereich M-Thetis. The card is not in the repo. Her question about an M-Thetis intercept or plaintext pair is unanswered. No finished search moves. Phase 78 records the desks now working. FRA has acknowledged the Swedish request. Heather Steele has been asked to photograph NARA RG 457 Box 620, NR 1665, and has not answered. The Bauer collection is intact in Duivendrecht; Selm will inspect it. ARGUS 10.1 states that a 1945 Nixe document carries `OKM 4/Skl IIc 140/45 g.Kdos.` and treats 4/SKL IIcc as a provisional producer of naval keys. The package forbids equating that office with M-Thetis and forbids restarting compute. No box has been read. Phase 79: Frode Weierud confirms that the allotment list in force on 1 May 1945 was Hering, replacing Haifisch on 4 March 1945, and that the bigram tables were Quelle. The undated 621–6ƃ3 fragment is still not identified as that edition. The scan he sent, Prüf-Nr. 3271, is Dorsch, Forelle, Haifisch and Zander, with 25 May 1943 distribution stamps, and its columns were not transcribed.

Selm-team hapax leads × right rings	**Clean negative.** 28/28 placements dead at the board, **0 raw stops over** 4.471×10^{11} **settings** at 38.0M/s, 0 physical, 0 quarantine. A verdict on 28 placements of the *recorded* ciphertext with the middle ring pinned A — not on the phrases generally. All three are second-net/Potsdam, so the imported-register objection stands
Same leads, post-gap δ=4 geometry	**Clean negative.** 28/28 dead, **0 raw stops over** 4.471×10^{11} **settings** at 37.8M/s, 0 physical, 0 quarantine — an identical settings count to the exact arm, since splicing preserves menu shape and moves only rotor coordinates. Both readings of the transcript are closed for these placements
Tolerance 1 on those menus	**Measured affordable, not yet run.** All **24** qualifying menus project **0** stops at a measured **7.7**× (∼19.7 h). That corrects the expectation that low-loop menus would inflate: a menu contradicting everywhere at tolerance 0 leaves nothing for tolerance to widen. With exact and post-gap both empty, a mis-read *letter* is the only edit hypothesis left for these phrases
Middle ring ≠ A	**Done under the VIII-fast prior, clean negative.** Placement 1/24 dead over 4.152×10^{11} settings at all 336 wheel orders; placements 2–24 **23/23 dead, 0 raw stops over** 1.194×10^{12} **settings** at 72.1M/s, 0 quarantine. Scope is 42 of 336 orders; the other 294 remain open. Control breaks at full middle × right coverage
Long menus (len 28–40)	Two maximum-contamination re-runs complete: Phase 16 body arm 40/40 dead with 0 raw stops; Phase 8 top-30 reproduces 73 raw stops with zero additional survivors and no break. The 300-place catalog remainder is retained but deprioritised; ≤25 letters unaffected
Ciphertext garble risk	Sister P1030681 needed degarbling; exact bars may drop true shells. **17 of 48** published U-534 keys fail to reproduce their own recorded plaintext, but from at least three causes — multi-part per-part keys, indels, and only **4 controls / 8 events** of isolated substitution. An earlier "16 controls prove archive garble" reading is **withdrawn**
Crib-free (Ostwald/Weierud)	Still negative at 72 letters, but current-model exhaustion is now measured. Unseeded generic: 5/30, median −0.2306. Single-plug exhaust-6 reaches −0.0672 (about 71 % of the deficit removed) and saturates by exhaust-10 at −0.0567. New two-plug exhaust-6 gives 30 % wins / −0.0361; widening to exhaust-10 gives 40 % but worsens the median to −0.0486. Thin controls, no zero crossing, no break
Tolerant board sensitivity	**Proven on a known key**: at 1–3 corrupted crib letters the exact board *loses* the true setting and a tolerance-≥ g board *keeps* it, forcing 25/25 plugs correctly. Metal port cross-checks at **0/192** lane mismatches

Tolerant board specificity	**Placement**-dependent, not crib-length dependent. An early "≥17 edges buys tolerance 1" rule held only at offset 0 and is **withdrawn**: 18-edge menus admit 4,301 and 16,019 survivors elsewhere. Tolerance amplifies an under-determined menu rather than rescuing it, so it is a per-menu decision
Tolerance 1 against P1030680	**Stopped after menu 8 of 24** (2026-09-27). The process is gone and the log has no footer. Menus 4–8 are dead, which repeats the August three, `NFUN@18`, `NFUN@21`, and `FUN@19`, and adds menus 5 and 8. The rank order moved, so this index skips `UNFU@23`, `FUN@22`, and `NFUUU@21`. Menus 9–24 are unrun. Log `logs/campaign-tolerance1-strongest-resume-2026-09-27.log`
Post-gap $\delta = 4$ against P1030680	**Partial / suspended at 144/237**; entries 1–110 and 112–144 are dead, while entry 111's 8 raw stops yield 0 valid ≤ 10-plug completions. Resume `--bombe-menus 0 --bombe-from 145`; middle ring remains A
Nazi Blaster 9000 (unified Verilog Mulein Future Bank)	**Operational preflight and bounded production stripe DONE; 0 BREAK gates**: known-key receipts agree across source RTL, post-Yosys RTL, clear Yosys-JSON, and cleartext Float TensorLUT. Runtime selected `BANK_LANES=4` at 276.035 complete receipts/s. The *Fahrenheit 261* historical 261-entry campaign executed only shell 0, identity Future 0, and settings $0 \le s < 256$: 16/16 chunks, 6,656 receipts, zero positives, and zero BREAK gates; the other 260 Futures did not run. The separate operational inventory contains 628 Futures (314 identity + 314 post-gap $\delta = 4$). Shell 0, setting 0, and all 628 Futures completed: **628/628 chunks and 16,328 receipts**; 15 repaired/non-exact candidates yielded 0 exact hits and 0 gates. The same shell, all Futures, and settings $1 \le s < 256$ then completed **10,048/10,048 chunks and 4,163,640 receipts**, retaining 4,184 host-verified physical candidates: 40 exact identity hits, 2,056 identity repairs, 41 exact post-gap hits, and 2,047 post-gap repairs. **0 BREAK gates; no key or decrypt**. Read-only adjudication now kills all 41 exact post-gap hits at joint ≤ 10-plug completion, independently matching the 40 exact identity deaths: all **81 exact candidates are split-menu ghosts**. Identity-family and gapped post-gap repairs were climbed (Phases 62–64): **38 then 824 then 4,101 crib BAD, NO BREAK**. Phase 64 best −3.0390 (IC 0.0544) versus a 12-sample floor of −3.4564 after 3.33 million climbs is not a near-miss. The 2026 Ostwald ranker/beam/all-settings path is built (Phase 65); the 72-letter ranker curve did not greet. Broader settings and shells remain open. Bounded, not global. Cleartext, not FHE or encrypted tick rate

Identity-family one-edge repairs × Ostwald	**Clean negative (Phases 62–64).** Phase 62: 38 single-component repairs. Phase 63 Metal: 824 identity-family ≥ 4-plug repairs including split-menu trees, 4.6M decrypts/s, all crib BAD, best −3.4797 (IC 0.0395) versus noise best −3.4564 (Δ −0.0233). Phase 64 CPU: 4,101 identity plus gapped post-gap, 3.33 million climbs, **NO BREAK**, best −3.0390 (IC 0.0544) crib BAD. Beating a 12-sample floor after that many climbs is not a near-miss. Phase 65 ranker/beam/all-settings is built; the full 72-letter ranker curve scored 5/30 wins (greedy) and 6/30 with beam 4, both with a negative median — not a zero crossing, not pointed at P1030680. Phase 66 4-plug hot-letter brute: 0/32 oracle cover, 1/30 wins, median −0.1633. The 164-million 4-plug alphabet climb (Phase 67) **greets** on locked P1030684 VYAA / index 385320: 313 waves, ∼37.8 min at 36.3M/s, crib-exact, IC 0.0657, tail −2.9671, 10 pairs, known 72-letter plaintext. Log `logs/ostwald-fourplug-p1030684-true-385320.log`. That is a known-key greeting, not a P1030680 decrypt. The wrong-setting ghost (AAAA / index 0) is crib BAD: ∼38.0 min at 36.2M/s, IC 0.0509, tail −3.0462, **NO BREAK** (log `logs/ostwald-fourplug-p1030684-wrong-0.log`). A tempered plugboard integral (Phase 69, 8 controls, 3 seeds) improves the margin and worsens scale-free separation on every seed: compression, not a filter.

Soft-band Welchman stops can quarantine into Stochastic seeds (`--hybrid-quarantine`). Synthetic P1030684 block-wipe and historical P1030681 first-draft controls grade that path (escalate recovers edit ceilings, not a new key). On P1030680 itself, UEBUNG and Regenbogen soft bands produced German-looking IC without survivors under the 80% bar — coincidence, not plaintext.

6.1 A clean negative the board had not earned

The Metal bombe tabulates one upper-involution table per distinct slow-wheel (l, m) state across a menu span, and capped that at four. On overflow it wrote a zero survivor mask — the code for *eliminated* — for a lane it had never tested. Pure stepping arithmetic over every span, all 56 naval (middle, right) rotor pairs and all 676 windows (`Scripts/max_upper_audit.py`) bounds how often that fired: never at or below 25 letters, then 0.29 % of lanes at 28, 1.14 % at 30 and 5.56 % at 40. The damage is concentrated in the curated long-menu arms rather than spread thinly. Phase 16 arm 2 — forty mid-message anchors whose zero raw stops closed the Übung-push hypothesis — is 40 letters throughout, so *all forty* placements were contaminated and that negative was unearned as recorded. It has since been re-run on the corrected kernel: 40/40 dead at the board, zero raw stops over 6.387×10^{11} settings, so the cap was not concealing a key and the hypothesis stays closed on evidence. Phase 8's curated top-30, contaminated on 20 of 30 placements, likewise re-runs to 73 raw stops over 4.79×10^{11} settings — identical to the archived figures — with best board IC 0.041 / tail −4.819 and no break. In both arms the previously untested lanes yielded *zero* additional survivors. Two archival notes belong with that: the scored-completion count on the top-30 moved $0 \to 35$ purely because the head-reading gate and quarantine's soft IC floor both postdate Phase 8, the raw-stop count being the coverage-sensitive figure; and the archived

top-30 log is truncated at menu 6 of 30 with no summary, so its quoted figures had never been on disk until this re-run supplied them. The remaining 300 contaminated catalog placements are retained but deprioritised: the two maximum-contamination arms independently produced zero additional survivors, while the strongest examples now also appear in the completed VIII-fast middle-ring arm. The Regenbogen queue is clean, and everything at 16–25 letters, including all of the Thetis-register work, is untouched and stands. The cap is now 8 and overflow reports *undecided*, which the host — which has no such cap — re-tests in full: a table size can no longer discard a key.

6.2 Unpinning the middle ring for 6.5–11.4×, not 26×

Before Phase 51.12, every target campaign pinned the middle Ringstellung at A, which is exact only while the middle wheel stays off its own notch in span. The gap is real (∼8–15 % of keys) and was priced at a flat 26×. It is cheaper, because notch tests depend on *window* position and are ring-free: if neither a lane's middle wheel nor that of its ring-A partner at window $m - \rho$ reaches a notch across the span, the middle wheel steps only on right-wheel notches — the same times in both trails — the left wheel never moves, and $\text{offset}_M = m - \rho$ *is* the partner's window. Same scramblers, same verdict. Ring A therefore runs in full and rings B–Z run only their notch-hitting lanes, and the union is complete coverage rather than a sample. Graded on the validated host board over 2.4×10^6 claimed-covered pairs and then through the kernel: zero verdict mismatches, and with ∼5 600–6 900 survivors per ring actually discarded, every discarded verdict is carried identically by its ring-A partner. Predicted cost is 6.5× a right-ring pass with a one-notch middle wheel and 11.4× with two. End-to-end on the blind control at full middle × right coverage — 908 544 shells, 4.152×10^{11} settings, the first run in this campaign with no residual ring gap — the break is recovered with all ten plugs in 4 307 s at 96.4 M/s: 26× the search space for 12.2× the time of the right-ring pass, because roughly 70 % of lanes exit after the trail test. Raw stops rise from 2 to 4, which is the mechanism working rather than a regression: with the middle ring unpinned the true machine has several ring-equivalent representations in the swept space. Measured basis is ∼72 minutes per placement at full 336 wheel orders, and the VIII-fast subspace cut that to ∼6.6 h for the remaining 23. Against P1030680 the arm is **complete under that prior**: placement 1 dead at the board over 4.152×10^{11} settings; placements 2–24 **23/23 dead, 0 raw stops over** 1.194×10^{12} **settings** at 72.1M/s, 0 quarantine. This is VIII-fast subspace coverage (42 of 336 orders), not wheel-order elimination, and not a decrypt.

Two sieves were graded and rejected. The historical ten-lead board bounds the plugs a menu has not yet forced ($26 - \text{determined} \geq 2(10 - \text{pairs})$); it is sound — the blind control still breaks — and useless, taking 847 434 raw stops to 847 414 (0.002 %) for a 4 % throughput loss, because it only bites on long strong menus that already die at the board. Separately, three quarantine flags this report and the ledger both document were never parsed, so the soft-tail arm recorded at −4.3 ran at the default −4.0.

6.3 The crib-free path, measured

Every arm above needs a probable word, and every crib in the catalog is imported register from other nets. It is also recurrence-filtered: the miner kept only phrases carried by two or more decrypts. An internal relay audit on 2026-08-16 had already measured that 99.0 % of distinct 16–40 letter corpus windows are one-carrier phrases, yielding 2,691,455 legal placements, and emitted a 24-placement fixture. That fixture was never run, however, and the gap remained absent from this report and the public chronology. Selm Merel Wenselaers (Historian / Curator, Amsterdam / Antwerp) joined the campaign team after independently flagging the recurrence filter and supplying three specific historical leads. Her legal-placement arithmetic reproduced exactly, and her two highlighted menus are the only single-component members of their tied

loop-count tier. Her work supplied the historical judgement that turned a dormant inventory into an actual experiment; HELUT's graded machinery turned that selection into bounded measurements. These are joint campaign findings, and the combination of archival research and mechanical cryptanalysis is the point rather than a division of credit. The correction matters for interpretation: the catalog is the register *repeated across* the corpus, not its whole register. It still does not supply Thetis plaintext, which is the actual missing ingredient.

Her reported Volksliste III process—demolition parties aboard while crews transferred to Neustadt—also imposed a constraint on the machinery rather than a new crib. If two strings are linked stages of one procedure, they must be allowed to compete *independently*, and they must not be concatenated into a sentence nobody is attested to have sent. A short phrase can be historically strong and mechanically unusable at once: `NEUSTADT` is eight letters with no single-component placement, so the campaign's own rules bar it from a solo sweep, and it therefore could not contribute at all. A *constellation* resolves that. It is one joint hypothesis over several independent anchors in which only the absolute $(\text{step}, \text{plaintext}, \text{ciphertext})$ board constraints are unioned; the anchors are never joined, and no plaintext is asserted in the gap between them. The builder is fail-closed, so a malformed, overlapping or self-enciphering anchor rejects the whole constellation rather than quietly leaving a weaker menu that still resembles the plan. Exactness became the conjunction over anchors, and the tail statistic scores only physically contiguous uncovered runs, so removing two anchors cannot manufacture an n-gram across the excision or across the gap. The independence test became constraint-disjointness, which tightens multi-menu confirmation rather than loosening it. Staged from this: `FFFTTTBLEIBTBESETZT` $\times$ `NEUSTADT`, 24 single-component rows of 27 edges and 7–9 loops, both strings repo-verbatim, topology cross-checked between an independent union-find and the board. The cost is 3.83×10^{11} settings, roughly 2.8 h.

Result: a clean negative. The arm ran to completion on 23 September 2026: 24 of 24 menus dead at the board, **zero** raw stops across 3.832×10^{11} settings at 42.5 M/s, no physically buildable board, and no soft-band candidate at all—the quarantine file was never written. Zero raw stops is a stronger shape than the campaign's usual "many stops, none physical": the board never produced a candidate to sieve. Under right-ring coverage with the middle ring pinned, no key places `FFFTTTBLEIBTBESETZT` *and* `NEUSTADT` at any of the tested alignments. This does not disturb the reported Volksliste III history; what it falsifies is the narrower proposition that both strings sit verbatim in this ciphertext at those offsets. The residual stays printed: a pinned middle ring leaves ∼12–23 % of keys untested across these 40–67 letter spans.

Unicity: the barrier is the estimator, not the information. This was worth computing and had not been. The full key—wheel order, Greek wheel, thin reflector, rings and position (4.151×10^{11} settings) together with the ten-plug board $\binom{26}{10\ \text{pairs}} = 26!/(6!\,10!\,2^{10}) \approx 1.507 \times 10^{14}$—carries about $6.3 \times 10^{25} \approx 2^{86}$, i.e. ≈86 bits. German redundancy at roughly 3.2 bits per letter puts 72 letters at ≈230 bits, so unicity distance is near **27 letters** and the target sits at about 2.7× that. Exactly one key yields German, by a margin of some 144 bits. Every negative in this report is therefore a failure of *search*, not a shortage of information: the answer provably exists inside the space being enumerated.

An independent implementation confirms that the estimator is what binds. A from-scratch Python M4, verified by reproducing every clean key in the corpus, combined with leave-one-out n-grams and *genuine* wrong-key decrypts, reproduces the standing negative: median margin -0.053 at 72 letters across orders 2–5, against the Swift pipeline's -0.036 to -0.123. Two unrelated implementations agreeing on a negative is stronger evidence than either alone.

One methodological trap is recorded because it nearly became a false positive. The first version of that probe reported a margin of $+1.14$ and a 100 % win rate at 72 letters, which would have overturned the central negative of this work. The defect was supplying the true

plugboard to the truth *and* to every decoy: with the correct board the true setting decrypts to clean German while wrong settings are noise, so any scorer separates them perfectly. It was grading against the answer key. Requiring the board to be hill-climbed symmetrically at every candidate moved the figure from +1.14 to −0.044. That single correction is the difficulty of the problem, and it is why partial exhaustion must always be applied to the truth and the decoys alike. The probe's scope limit is equally worth stating: it trained on 4,644 letters, whereas the dense trigram table used here derives from 28.5 M. It therefore measures saturation in n-gram *order* at small sample size, not the exhaustion of language modelling as a lever.

The practical consequence is a change in ranking. The plugboard was a *daily* setting, so a second Thetis signal from 1 May would constrain the same 47-bit board with 144 letters rather than 72, and the oracle ladder already shows four correct plugs flipping the margin positive. Archival discovery therefore outranks further sweeping, including the five-to-seven day swept version of the arm above.

Register fidelity, not volume, is the language axis. The probe's small-sample caveat was tested directly rather than left standing. The corpus was widened beyond the U-534 pages—a 37 % increase in naval training text, 1,704 validated letters—and discrimination got *worse* on both metrics, on every seed, with the scale-free z ranges disjoint (mean z +0.135 → −0.527). The cause is register rather than quantity: ten of the twelve added messages are Swedish intercepts of *Luftwaffe* traffic (`TAETIGKEITSBERIQT`, `LFLXKDO`, `JAEGER`), against a Kriegsmarine naval-procedure target, and even the orthography differs (`Q` for `CH`). Diluting naval register with air-force paperwork degraded the model. So the language hypothesis is narrowed rather than refuted: fidelity to the target's own register is the axis, and volume of German is not. The corollary matters more for the archival arm than the mechanical one—a hundred letters of genuine Baltic training procedure outweighs ten thousand letters of Luftwaffe reports.

Three classical methods, closed by measurement. With the estimator and data routes closed, the remaining question was whether any human method had gone unapplied. *Cillies*: across the 47 recovered message keys, 11 carry a repeated letter against a uniform expectation of 10.2, and per-position frequencies are flat. These operators randomised properly, so there is no habit prior—now measured rather than assumed. *Indicator/Grundstellung*: the message key is a bijective function of the daily key and the Grund, so substituting one unknown for the other reduces nothing, and the 48 (indicator, message-key) pairs cannot be leveraged. *Banburismus*: requires depth, and the target has no partner.

One real lever: rotor VIII on the fast wheel. Every daily key recovered from 1 May 1945 agrees across three separate nets—438/AACU (31 messages) IV/III/VIII, 568/AAEL (16) V/VI/VIII, 528/AAFB (1) V/II/VIII. Under independent per-net choice that is $(1/8)^2 \approx 1.6\,\%$. This is tighter than the general convention prior already carried (one of VI/VII/VIII on the fast wheel, 126 of 336 orders) and is measured on this corpus rather than asserted: VIII specifically, **42 of 336**. The effect on the previously parked middle-ring arm, read from the sweep banner, is 113,568 shells per menu instead of 908,544, i.e. ≈6.6 h instead of ≈28 h, with `no residual ring gap` still printed so middle and right ring coverage is preserved. That arm is now **complete and empty**: 24/24 dead at the board, 0 raw stops on the VIII-fast slice, 0 quarantine. A negative here is not wheel-order coverage; the other 294 orders remain open.

It *eliminates nothing*. The prior reorders the search so the likeliest wheel orders run first; the remaining 294 stay open, and a negative under it is not wheel-order coverage. Three nets is a thin sample, the nets may have drawn from a shared Schlüsseltafel rather than choosing freely, and these are the keys that happened to be broken. Whether 1 May allocation assigned fast wheels centrally across nets is a historical question and is referred to the archival arm:

documented central assignment would make this a strong constraint on Thetis, while free per-net choice leaves it a weak ordering hint.

The measurement that matters is not the price but the geometry, and it sets the run order. A 27-edge contiguous crib spans 27 positions; these constellations span 47–67, because the anchors sit far apart. Span, not edge count, is the quantity a middle ring pinned at A fails to cover: the engine's preflight reports ∼12 % of keys uncovered for one middle-wheel notch and ∼23 % for two across a 67-letter span, against a few percent for a compact crib. Loops are not free: the extra deduction power a joint menu buys is partly paid for in a wider blind spot, which is the residual quoted with the result above. The Ostwald–Weierud ciphertext-only attack needs none: for each candidate rotor setting it hill-climbs the plugboard and ranks the setting by the score the climb reaches. Its published reach is messages down to roughly 100 letters, with 78 the shortest broken, on three-rotor Heer traffic; P1030680 is 72 letters on four-rotor naval M4, i.e. shorter than the record on a harder machine.

The limit is not throughput, which changes how hardware should be spent. Seventy-two letters of naval German carry ∼223 bits of redundancy, a ten-plug board is ∼47 bits of nuisance parameter *fitted per candidate*, and the shell×position space is $\sim 10^{11}$: taking a maximum over that many overfitted candidates is a multiple-comparisons problem, and adding candidates raises the maximum of the noise. The decisive quantity is therefore the *margin* — climbed score at the true setting minus the best climbed score over wrong settings — and `--ostwald-curve` measures it at a ladder of truncation lengths over the published 1 May 1945 U-534 keys, with exhaustion applied symmetrically to true and decoy settings so the margin cannot be manufactured.

The harness refuses to report a curve unless its controls decrypt their own published plaintext under their own published key, and that preflight found something: **32 of 48 controls round-trip exactly**, the rest do not. An earlier revision read that as transcription garble corroborating the garble hypothesis; **that reading is withdrawn**. Classified by error shape, the 17 discrepant controls have at least three causes. Most are *multi-part* messages — several exceed the ∼320-letter Kriegsmarine limit and were sent in parts with a fresh Grundstellung each, so end-to-end decryption under one message key correctly yields a good head and a garbage tail, which is precisely the observed shape. A few were first read as indels or divergent transcripts. The short-message exemplar was classified on the glued scrape and is not re-established (Phase 74). Only four controls, eight events in total, are isolated substitutions — far too thin to fit a confusion model, so any degarbler must take its prior from Girard's direct reading of the originals (he names U↔N, Q↔G, H↔F) rather than from this corpus.

Validation is unambiguous — at 372 letters the staged climb recovers all ten plugs and 100 % of the plaintext (margin +1.64, $z = 35.6$) — which is what licenses believing the negatives. Staged scoring (IC, then bigrams, then trigrams, matching the reference implementation's default) decisively beats the bigram-only climb previously in the repo: at 252 letters, 8/10 plugs against 0/10. The measured threshold is ∼200 **letters**, where the median margin crosses zero; control counts above 180 are thin ($n = 3$–6), so those medians are noisy while the monotone trend is not. Ostwald's partial exhaustion — fixing one plug and climbing the rest over all 141 candidates — removes **68 %** of the deficit at 72 letters ($-0.2389 \to -0.0754$ on identical controls) and doubles the win rate, the largest single lever found and the one the literature names.

Unseeded, the margin at 72 letters remains *negative*: the crib-free path is not yet a viable standalone arm against P1030680, our threshold is 2.8× worse than the published record, and that gap is unclosed.

The lever available only here changes that picture at the target length. A Welchman stop arrives with the stecker its menu forced — 15–24 of 26 letters, i.e. 7–12 plugs — and the reference implementation documents exactly that interface (`-s`, "known plugboard connections, e.g. from running a bombe") while requiring it to be fed by hand, because the bombe is someone else's program. Modelled fairly, with the true setting seeded from k correct plugs and every decoy seeded from k *random* plugs, at 72 letters on ten controls the margin crosses zero at $k = 4$

(+0.130) and at $k = 8$ the attack recovers the complete key and 100 % of the plaintext on every control (+1.049, 10/10 plugs). That is below the published 78-letter record, on the harder machine — but it is *oracle-seeded*, so it assumes the seeded stop is the true key. What it establishes is narrower and still useful: a *true* bombe stop is finishable at 72 letters, so a stop the strict linguistic bar rejected for garble or short-message noise can be recovered.

That makes the climber a much better escalator than the genetic path of §Quarantine, which returned zero survivors at a ∼60 % coincidence ceiling without saying why. Escalating every soft-band stop the campaign has ever quarantined — 55 UEBUNG, 22 Regenbogen scuttle, 7 Hela, 2 from the Phase-49 re-run, 86 in total — yields no break, and in every arm the best stop scores *below* the best of twelve random settings (Δ from -0.19 to -0.80). The soft band is not a near miss; it is worse than noise under a scorer that would finish a true stop. The quarantine escalation path is therefore closed with a reason rather than a ceiling.

6.4 A diagonal board that counts contradictions

On the physical bombe a contradiction *is* current finding a second path through the diagonal board: the hypothesis short-circuits and the drum advances. Copper cannot answer "how many contradictions?", and every faithful reimplementation inherits that, including ours, which returns `nil` on the first doubled row. Under garble this is fatal — one mis-transcribed ciphertext letter contradicts a *true* menu — so every exact-crib clean negative above is a negative about the *recorded* ciphertext rather than the transmitted one.

In silicon the question can be posed differently: how few menu edges must be deleted before a setting closes consistently? Accepting a setting when $\leq t$ deletions suffice gives a *deletion-tolerant* diagonal board, occupying the structural slot Welchman's diagonal board did — a new board on the bombe, not a new bombe. Tolerance 0 is the historical board bit for bit.

Tolerance must *remove* edges before propagating rather than abandon them mid-flight. A first implementation did the latter and measured as an exact no-op, for two compounding reasons worth recording: by the time a conflict surfaces the bad value is already committed and has propagated through the diagonal board, and the edge visited when it surfaces is typically a *correct* edge rather than the garbled one, so blaming it leaves the real culprit in place.

Sensitivity, graded on the published P1030684 key with letters corrupted inside a 27-letter crib span: at 1, 2 and 3 garbles the exact board **loses** the true setting and a tolerance-$\geq g$ board **keeps** it, while still forcing 25/25 plug deductions correctly. Since a true stop needs only four correct plugs to flip the crib-free margin at 72 letters, the two mechanisms compose: a garbled true key survives the board *and* is finishable. The Metal port is graded against the host board rather than trusted, at **0/192** lane mismatches across 45 crib × offset × tolerance cells, with wall time per shell rising 0.03 s → 7 s from tolerance 0 to 2 — confirming the extra closures are performed and not elided.

The cost is survivor inflation, and here an early reading was wrong and is withdrawn. At crib offset 0 inflation looked like a clean function of menu size: 16 edges admitted 130,787 of 456,976 lanes at tolerance 1, while ≥ 17 edges admitted exactly one, and ≥ 22 admitted one even at tolerance 2. Varying the *placement* destroys that rule: 16 edges costs nothing at offset 15, while 18-edge menus admit 4,301 and 16,019 survivors at offsets 40 and 65. Crib length does not predict inflation; the menu's loop structure does, because tolerance spends redundancy and only a loop-rich menu has enough to spend. The placements that detonate are those already weak at tolerance 0 — tolerance *amplifies* an under-determined menu rather than rescuing it. Tolerance is therefore a per-menu decision, pre-qualified by measuring inflation against the escalator's budget, and not a global switch.

The known-key sensitivity grade does not assert that P1030680 *is* garbled. The target tolerance-1 arm has a bounded August prefix: **3/24** strongest placements of that day's order are dead at the board (`NFUN@18`, `NFUN@21`, `FUN@19`). The 2026-09-27 resume stopped after menu 8 of 24. The process is gone and the log has no footer. Menus 4–8 are dead. The rank order now sorts

equal-loop menus by span, so that index skips `UNFU@23`, `FUN@22`, and `NFUUU@21` and repeats the three negatives. Menus 9–24 are unrun. Log `logs/campaign-tolerance1-strongest-resume-2026-09-27.log` This is not a verdict on garble generally.

Indels are the board's *second* mechanism rather than an omission: Girard records an entire four-letter group (`HMHY`) present on one transcript and absent from the other. That frame shift is handled by re-indexing the menu and shifting step numbers — a geometry change on the exact board, with no survivor inflation or pre-qualification gate. Against P1030680, the post-gap $\delta = 4$ arm is **suspended at 144/237**: entries 1–110 and 112–144 are dead; entry 111's 8 raw stops produce no valid ≤ 10-plug completion. Resume `--bombe-menus 0 --bombe-from 145`; 93 hypotheses remain and the middle ring is still A. No physical candidate or break has resulted.

A production Verilog Future Bank unifies explicit Future geometry, exact-first one-edge repair, a shared 80×26 trail, and held receipts. On P1030684 controls, complete receipts agree across source RTL, post-Yosys RTL, clear Yosys-JSON simulation, and cleartext Float TensorLUT, including held-valid backpressure.

Identical complete-receipt benchmarks then compared bank widths 1/2/4/8/16 on 16 settings $\times$ 16 jobs $\times$ five repetitions. All widths produced digest `fnv1a64-17435188996435da`; median rates were 233.777350172, 264.718510307, **276.035471632**, 273.889660462, and 230.780129610 receipts/s. Thus `BANK_LANES=4` was selected by runtime rather than synthesis size. Its graph has 189,032 LUT6, 2,772 DFF, 205,177 wires, and 28 levels; the selection receipt is `logs/mulein-future-tensorlut-selection.json`.

The protocol-v3 target runner binds one immutable manifest byte snapshot, the netlist, graph, width, protocol, and exact shell/Future/setting plan into its run identity. Resume admits only an exact ordered plan prefix. Every synchronized chunk stores the canonical per-job receipt projection, permitting digest recomputation and derivation of hit and BREAK-gate state rather than trust in summary fields. Persisted hits are host/scoring-replayed; a prior valid gate halts before evaluator submission. Semantic row drift, sparse or reordered chunks, candidate deletion, and records after a gate fail closed. Atomic non-truncating open and a retained exclusive lock enforce one writer through final synchronization. Metal status, tags, seeds, held receipts, consume/rearm, and timeout are also checked. Only exact identity/no-drop positives may enter the existing discriminator; repaired and post-gap positives remain non-BREAK-eligible until their correction or full-message geometry is independently replayed.

The historical *Fahrenheit 261* campaign stages 261 menu/Future entries, but its bounded launch covered only shell 0 `B/beta/IV-III-VIII/AAAA`, identity Future 0, and settings $0 \leq s < 256$. Protocol-v3 run `sha256-e6dc10d4...2fc1360` completed **16/16 synchronized chunks and 6,656/6,656 canonical receipt projections, with zero hardware positives and zero BREAK gates**; identical resume revalidated all 16 rows and appended nothing. Durable receipt: `logs/p1030680-mulein-unified-smoke-v3.jsonl`, SHA-256 `687d0838...2b1f7e7d`. This remains one clean historical local negative, not coverage of that manifest's other 260 Futures.

Nazi Blaster 9000, the formal name of the unified P1030680 Mulein search machine and overall campaign, then introduced a separate operational/scuttle inventory rather than rewriting the Fahrenheit 261 artifact: 628 Futures (314 identity + 314 post-gap $\delta = 4$), fingerprint `fnv1a64-616326e94036a97d`. Shell 0, setting 0, and all 628 Futures completed **628/628 chunks and 16,328 checked receipts**. Fifteen host-replayed physical candidates survived—5 identity-family one-edge repairs and 10 post-gap one-edge repairs—but every candidate was non-exact. Therefore the grade is **0 exact hits and 0 BREAK gates**, not "zero positives." Repaired candidates remain non-BREAK-eligible without explicit correction replay; post-gap candidates require geometry-aware full-message replay. They do not establish that the ciphertext is garbled. The durable receipt is `logs/p1030680-mulein-operational-preflight-v3.jsonl`, SHA-256 `55a68266...9a17f993`.

The bounded production stripe is **DONE as bounded coverage**: the same shell, all 628 operational Futures, and settings $1 \leq s < 256$. From `2026-08-20T20:44:18Z` through

2026-08-21T01:54:45Z, protocol-v3 run sha256-efb770bc...592cc0 wrote **10,048/10,048 new synchronized chunks, 0 skipped, and 4,163,640/4,163,640 canonical receipt projections**. The durable receipt is the settings-000001-000256-v3 production ledger under logs/; its SHA-256 is c516238d...b1e7e52f2.

Host replay retained **4,184 physical candidates**, all verified: 40 exact/no-drop identity hits killed by $\leq$ 10-plug completion, 2,056 one-edge identity repairs, 41 exact/no-drop post-gap $\delta = 4$ hits, and 2,047 one-edge post-gap repairs. There are **0 row BREAK gates, 0 candidate BREAK gates, no scored plaintext, no key, and no decrypt**. This is not a zero-positive result or a clean global negative. A translation audit paired 1,724 identity/post-gap candidates at a four-setting-lane shift; over 82% of each family therefore resamples the same closure/live state in this stripe. That result does not eliminate a missing group at another physical boundary, but makes blind widening and same-shape leading $\delta = 6/8$ stripes low-information.

The GPU arm that just completed is the four-placement P1030668 BLEIBTBESETZT context fixture under --subspace viii-fast-wheel --bombe-middle-ring --bombe-ring-sweep (Phase 60; log campaign-wenselaers-bleibt-besetzt-viii-fast-middlering.log): **4/4 dead, 0 raw stops over** 2.076×10^{11} **settings** at 52.3M/s, 0 quarantine. Live archival lines outrank another GPU tranche (Phase 61). Host-side, identity-family Ostwald against the bounded stripe is now a clean negative three times (--mulein-ostwald-adapt, then --ostwald-escalate): Phase 62 climbed 38 single-component repairs (**NO BREAK**, best -3.6976 versus a random-setting floor of -3.4564); Phase 63 Metal-climbed **824** identity-family $\geq$ 4-plug repairs including split-menu trees (**NO BREAK**, best -3.4797, Δ -0.0233, 4.6M decrypts/s, 8 GB unified); Phase 64 CPU-climbed **4,101** identity plus gapped post-gap repairs (3.33 million climbs, **NO BREAK**, best -3.0390, IC 0.0544, crib BAD). The Phase 64 Δ versus a 12-sample floor is multiple comparisons, not a near-miss. The 2026 Ostwald ranker, beam climb, and no-IC-sieve all-settings runner are built (Phase 65). The full 72-letter ranker curve (--ostwald-scorer ranker) scored 5/30 wins greedy and 6/30 with beam 4, both with a negative median: not a zero crossing, and not pointed at P1030680. Fitting against climbed decoys made it worse (1/30). A 4-plug hot-letter brute at exhaust 8 has 0/32 oracle cover: the eight hottest ciphertext letters do not contain four true plugs. The 164-million 4-plug alphabet climb (Phase 67) **greets** on locked P1030684 VYAA / index 385320 (log ostwald-fourplug-p1030684-true-385320.log): crib-exact, IC 0.0657, known 72-letter plaintext. That is a known-key control, not a unique 72-letter IC filter, and is not a P1030680 decrypt. The wrong-setting ghost (AAAA / index 0) is crib BAD (IC 0.0509, NO BREAK; log ostwald-fourplug-p1030684-wrong-0.log). The 40 exact identity hits need no rerun because completion already killed them. Operational/scuttle hypotheses ran first as a working search prior, not historical proof. Settings $256 \leq s < 456{,}976$, other shells, and the catalog, middle-ring, tolerance, and legacy-indel remainders remain open. The rates, receipts, and fitness are cleartext Float TensorLUT/Metal—not encrypted tick rate, not FHE, not a key, and not a decrypt.

7 Parallel Track: TensorLUT

Independently of the Boolean catalog, HELUT ships a continuous–to–discrete compiler: Yosys LUT6 cells become 64-wide floating INIT tensors, Metal evaluates a multilinear stream, λ squeezes toward binary, and an emitter writes gate-level Verilog. Unmutated enigma_m4 locks $F_{\text{crypto}} = 0$ (925 LUT6 + 49 DFFs baseline). Full INIT cold-start and stecker-cone melts *shatter* under λ; the live arm freezes known-good silicon and evolves a $\leq$ 10-pair stecker involution by construction (blind three-pair rediscovery PASS; claims C8–C9). Six formal lemmas of the continuous–discrete loop hold in-repo (claim C19; directives/tensorlut-theorem.md). Melt–freeze–snap on a separable Boolean interpolant is claim C44; a 2-LUT cascade melt that emits Verilog and matches 8 Boolean corners is claim C48 (dual interpolants allowed). This grades generative hardware synthesis and reciprocal genotype search. It does **not** invent P1030680's

missing plaintext (N7, H6).

8 Scope Relative to the HELUT Stack

HELUT also graduates a torus FHE path (LWE/GLWE, GGSW blind-rotate, certificates) documented in `paper/helut.tex` and `directives/claim-sheet.md`. That path is out of scope for campaign fitness: P1030680 search remains cleartext Metal batch. Encrypted SING envelopes and hardness calibration are stack claims (C4–C7, H1–H3), not evidence that this message is broken.

9 Conclusion

P1030680 remains unbroken. The contribution of this campaign is a graded, reproducible Boolean and stochastic laboratory on Apple Silicon; archival elimination of wrong nets and many wrong wedges; scoring and ghost-control fixes that make short-M4 search honest; the benchmark-selected *Nazi Blaster 9000* unified Verilog-to-TensorLUT machine; and durable bounded receipts for the *Fahrenheit 261* historical slice, the all-Future operational preflight, and the completed settings-1 . . . 255 production stripe. That stripe retained 4,184 host-verified physical candidates but fired zero BREAK gates and supplied no scored plaintext, key, or decrypt; repaired and post-gap candidates remain outside the gate pending explicit replay. A parallel TensorLUT arm learns reciprocal structure without claiming a decrypt. Geometry-aware candidate replay is now complete for every *exact* candidate, and it corrected this campaign's own diagnosis. Counting the durable ledger rather than quoting it gave **41** exact/no-drop post-gap candidates, **40** exact identity candidates already killed by plug completion, and **4,103** repaired. A read-only adjudicator then graded the 41 with no GPU, because the ledger already persists shell, setting, seed, Future identity, and repair provenance; it writes its own ledger, since the campaign ledger re-derives every candidate and compares by equality, so mutation would invalidate all 4,184 rows. Host replay verified **41/41**, and **all 41 return** `no-consistent-board`: no $\leq$10-plug involution satisfies the whole menu, so they never reach the language bar and the missing geometry-aware decrypt was never the binding constraint. Their six source Futures carry **two to five components** on 16–21 edges, one a zero-loop tree, making them split-menu ghosts of the shape documented since Phase 4 — and the campaign had already killed all 40 identity siblings by the same sieve through an independent path. **All 81 exact candidates in that stripe are ghosts.** A P1030684 calibration control gates the scorer and additionally proves the sparse order-3 mean reproduces the dense trigram score *bit-exactly* on hole-free geometry, so the -3.600 / 0.055 bars transfer in scale after all; what does not transfer is the dense crib-excising *sample*. The **4,103 repaired candidates are deferred**: a repaired hit is crib-exact-minus-k by definition, so they need a declared bar and human sign-off rather than more code. The actionable lesson is about inventory, not compute: the next manifest should gate on connectivity rather than crib length, which would have excluded all six of those Futures before a single setting ran. The completed stripe is bounded evidence, not a global negative; broader settings, shells, the remaining 294 wheel orders, and the printed legacy remainders remain open.

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Frode Weierud and Olaf Ostwald's published ciphertext-only work is the basis of the crib-free arm measured above. The U-534 corpus, the transcriptions, and the sister-message degarbling that motivates the tolerant board are the work of Michael Hörenberg and Jean-François Girard; this campaign is downstream of their archival effort and would not exist without it.

Selm Merel Wenselaers (Historian / Curator, Amsterdam / Antwerp), a deeply valued member of the campaign team, joined the work after independently identifying the recurrence filter and supplying three verbatim-attested, historically motivated one-carrier leads. Her 15-of-44 legal-placement calculation reproduced exactly; her two highlighted placements were also the only single-component menus among the four tied at the highest loop count. Her archival judgement moved a dormant internal surface into an actual campaign and public view. The resulting work is joint: Selm's historical and archival research defines evidence-backed questions, and HELUT's graded machinery turns them into controlled measurements. The campaign is stronger for combining both disciplines, and the findings belong to that collaboration.

For chronology, HELUT's `Scripts/relay_crib_mine.py` had enumerated the general 99.0 % hapax surface on 2026-08-16, but its 24-placement fixture had never been run or reported. Selm's independent diagnosis, exact leads, placement arithmetic, and menu-selection insight supplied the historical prioritisation that made the surface actionable. She now leads the campaign's historical and archival arm, including primary-source and collection enquiries, while remaining part of the decision loop before new computational tranches begin.

Christoffer Duif (Esbjerg) made contact through Frode and Selm and is a potential collaborator on the cryptanalytic side, with his own GPU software bombe and a reported body of roughly 2,300 tried crib positions; Dan Girard, who worked on P1030680 years earlier, has shared an older list of attempted cribs with him. Their negative-coverage maps are source-reported and not yet reconciled with HELUT's graded arms, so no result above rests on them.

Naming collaborators records provenance; it does not imply that any person named above has claimed a decrypt of P1030680.